

Hierarchical Insider Threat Detection: Integrating Lightweight Screening with LLM-based Chain-of-Thought Reasoning and Knowledge Graphs

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Abstract—Insider threats pose a significant challenge to organizational cybersecurity, as malicious actions are often concealed within normal user workflows and easily evade traditional statistical or rule-based methods. To address the limitations of prior feature-based and deep learning approaches, this work proposes a hierarchical, two-stage insider threat detection framework that explicitly integrates cost-aware filtering with structured semantic reasoning. In the first stage, lightweight supervised machine learning models efficiently screen out the most suspicious behavior sequences, substantially reducing the computational bottleneck for subsequent analysis. In the second stage, these high-risk sequences undergo fine-grained analysis using Large Language Models (LLMs) augmented with Chain-of-Thought (CoT) prompting and knowledge graph integration. This combination enables the robust detection of complex, multi-stage, and cross-dimensional attacks. Supported by comprehensive evaluation metrics, ablation studies, and robustness analysis under data drift scenarios, the proposed approach achieves high detection performance on highly imbalanced datasets while providing interpretable, human-readable insights to support informed security operations.

Index Terms—Insider Threat Detection, Large Language Models, Chain-of-Thought, Knowledge Graphs

I. INTRODUCTION

With the rapid expansion of organizational information systems, insider threats have become one of the most challenging problems in cybersecurity [1]. Insider threats are typically initiated by legitimate users who may act maliciously or unintentionally cause security risks through negligence. Compared to external attacks, insider threats are significantly harder to detect because malicious actions are often interwoven with normal user activities, making it difficult for traditional rule-based or statistical methods to distinguish between benign and harmful behaviors.

Existing insider threat detection systems primarily rely on structured log analysis and supervised machine learning models [2]. These approaches aggregate user behaviors into statistical features and apply classifiers to identify suspicious users. While early research introduced graph-based mining to capture interactions between entities [3], the field has more

recently evolved toward deep representation learning to model increasingly complex behavioral sequences [4]. However, as attack patterns become more sophisticated, there is a growing need for models that provide not just a detection score, but also the semantic reasoning behind an alert. Large Language Models (LLMs) offer a potential solution for this type of reasoning [5]. However, the real-world challenge is balancing this reasoning depth with its operational cost.

II. PROBLEM STATEMENT

Although supervised machine learning approaches have been widely applied to insider threat detection [2], [3], traditional models primarily rely on manually engineered structured features and lack the ability to reason over complex semantic relationships and long-range contextual dependencies. As a result, they struggle to capture multi-step and cross-dimensional attack patterns, such as the progression from sensitive file access to compression and eventual data exfiltration [4].

On the other hand, while LLMs demonstrate strong capabilities in semantic understanding and reasoning [5], directly applying LLMs to massive log data presents two major challenges [6]:

- High computational cost.
- Context length limitations that restrict the processing of long behavioral sequences.

Therefore, achieving high detection performance while controlling computational cost and effectively leveraging the reasoning power of LLMs becomes the central problem addressed in this study.

III. PROJECT DESCRIPTION

A. Motivation and Practical Relevance

Insider threats remain a high-impact cybersecurity risk because the attacker is a legitimate user whose actions can closely resemble benign workflows, making detection difficult with rule-based monitoring alone [1]. In operational environments, security teams require not only accurate alerts but also actionable explanations (e.g., which behavioral transitions

triggered suspicion and how events relate across systems) [8]. This need is amplified by two practical constraints: (i) enterprise logs are large-scale and heterogeneous (login, file, removable media, email), and (ii) malicious cases are typically rare, leading to strong class imbalance and a high cost of false negatives.

Recent advances in representation learning have improved modeling capacity for sequential behavior, yet many solutions still behave as black-box predictors and provide limited semantic justification for analysts [4]. In parallel, Large Language Models (LLMs) show strong reasoning and explanation capabilities when guided by structured prompting (e.g., Chain-of-Thought) [5], and surveys indicate increasing interest in using LLMs for cyber threat detection [6]. However, direct LLM processing over massive log streams is often computationally prohibitive and constrained by context-length limitations [6], [9]. These factors motivate a design that can preserve detection sensitivity while controlling inference cost and improving interpretability.

B. Problem Definition and Scope

We address the following problem: given multi-source system logs representing user activity sequences, detect insider threat behaviors (binary classification) and provide interpretable rationale for the decision. The target setting matches widely used insider-threat benchmarks where benign activities dominate and malicious behaviors manifest as multi-stage patterns spanning heterogeneous event types (e.g., sensitive file access followed by staging and exfiltration) [2], [4]. The scope of this project focuses on (i) offline modeling on a benchmark dataset for reproducible evaluation, and (ii) robustness analysis under distribution shifts that emulate evolving insider behaviors.

C. Research Objectives and Testable Claims

This project is guided by three objectives:

- **O1 (Efficiency–Recall Trade-off):** Reduce the computational burden of semantic reasoning while preserving high recall for rare malicious cases.
- **O2 (Multi-stage Pattern Detection):** Improve detection of cross-event, multi-step insider behaviors that are difficult to capture with purely aggregated statistics.
- **O3 (Interpretability for Analysts):** Produce human-readable explanations that connect suspicious events and their relational context.

Correspondingly, we formulate testable claims:

- **C1:** A hierarchical design (with lightweight screening followed by LLM-based reasoning) can achieve comparable or improved detection metrics versus feature-only baselines with substantially lower LLM inference cost.
- **C2:** Chain-of-Thought prompting improves detection of multi-stage insider behaviors by enforcing stepwise reasoning over temporally ordered narratives [5].
- **C3:** Incorporating relational context (via knowledge-graph summaries) improves robustness in cross-

dimensional attack patterns compared to text-only prompts.

D. Proposed Approach Overview

We propose a two-stage framework. Stage 1 uses supervised models trained on structured statistical features derived from log aggregation to rank risk across all user sequences [2]. The design objective is *recall preservation* under class imbalance: we intentionally prioritize retaining suspicious candidates, forwarding only the top $k\%$ sequences to Stage 2 for deeper reasoning. Stage 2 converts heterogeneous logs into standardized natural-language narratives and applies LLM-based semantic analysis with Chain-of-Thought prompting [5], [8]. To further support structured reasoning, we represent events as entity–relation structures (users, files, devices, accounts) and provide knowledge-graph-derived relational summaries to the LLM, enabling it to reason over cross-entity dependencies rather than isolated events [3].

E. Evaluation Plan and Reproducibility

Because insider threat detection is typically imbalanced, we evaluate using metrics that reflect rare-class performance, emphasizing precision–recall trade-offs and PR-AUC [6]. We will report both detection performance and efficiency indicators (e.g., fraction of samples forwarded to Stage 2 and relative inference cost). To ensure reproducibility, we will document preprocessing, feature construction, prompt templates, and ablation settings, and we will conduct controlled ablations to isolate the contributions of hierarchy, Chain-of-Thought prompting, and knowledge-graph augmentation.

IV. STATE-OF-THE-ART REVIEW

This section reviews insider threat detection research across four methodological lines: feature-based supervised learning, graph-based modeling, deep sequential representation learning, and emerging LLM-based semantic reasoning. It also identifies the gaps that motivate the proposed hierarchical framework.

A. Feature-based Supervised Detection

A large body of insider threat detection work relies on supervised models trained over aggregated statistics computed from structured logs [2]. Typical pipelines transform heterogeneous events into user-level or session-level features (counts, ratios, temporal summaries) and apply classifiers such as tree ensembles to output risk scores. These methods are attractive for scalability and operational simplicity. However, their reliance on handcrafted aggregation can discard ordering and context, limiting sensitivity to multi-stage insider strategies where the *sequence* and *transition logic* matter (e.g., gradual staging before exfiltration) [4]. In addition, feature-based models often provide limited semantic explanation beyond feature importances, which may not be actionable for incident response.

B. Graph Mining and Relational Modeling

Graph-based approaches aim to capture interactions among entities (users, files, devices, communications) and detect suspicious substructures or anomalous relational patterns [3]. By modeling entity dependencies explicitly, graphs can represent cross-dimensional behaviors that are difficult to express via flat feature vectors. Nevertheless, classical graph mining solutions may require careful definition of nodes/edges and can be sensitive to design choices (e.g., time granularity, edge semantics). Moreover, many graph methods prioritize detection scores rather than analyst-facing explanations and may still struggle with evolving attack patterns without frequent model updates.

C. Deep Representation Learning for Behavioral Sequences

More recent research emphasizes deep learning to encode complex behavior sequences and long-range temporal dependencies [4]. Sequence models can in principle capture multi-step patterns more effectively than static aggregation. However, two limitations remain central. First, deep models often behave as black boxes, which reduces interpretability in security operations. Second, training and deployment can be costly, and generalization under distribution shift (new roles, new tools, new policies) remains challenging in practice. These issues motivate complementary mechanisms that explicitly support reasoning and explanation.

D. LLM-based Reasoning and Explainability

LLMs provide a different capability profile: rather than learning solely from numeric features, they can operate on natural-language descriptions and produce step-by-step rationales. Chain-of-Thought prompting is a widely adopted technique to induce structured reasoning by forcing intermediate steps before the final decision [5]. For cybersecurity applications, recent surveys suggest LLMs can enhance semantic understanding and produce interpretable outputs for threat detection tasks [6]. Recent work on LLM-based log analysis also shows that prompt strategies can improve both performance and interpretability in log-related tasks [8]. However, applying LLMs directly to massive logs faces two structural constraints: (i) high computational cost, and (ii) context-length limitations for long sequences [6]. This indicates a need for architectures that selectively allocate LLM reasoning to only the most informative/suspicious segments.

E. Why a Hierarchical Design is Needed

Across the above lines of work, a recurring trade-off emerges: scalable feature-based models lack deep semantic reasoning, while LLM-based reasoning is semantically rich but expensive to apply at scale [6], [9]. A hierarchical design provides a principled middle ground: a lightweight stage screens and ranks candidates with high recall, and a reasoning stage performs fine-grained semantic analysis on a constrained subset. This strategy is particularly aligned with insider-threat settings where false negatives are costly and malicious instances are rare. In addition, hierarchical designs enable systematic cost–performance trade-off studies by tuning

the screening proportion ($k\%$) and measuring both detection performance and inference cost.

F. Knowledge Graphs as Structured Context for Reasoning

Knowledge graphs can provide structured relational context that complements text narratives, representing entities and their interactions as explicit relations [3]. In insider detection, relational modeling is useful for capturing cross-entity dependencies (e.g., a user accessing a sensitive file, compressing it, copying to a removable device, then contacting an external account). When used to generate relational summaries for prompts, knowledge graphs may constrain reasoning to grounded event relationships and potentially improve robustness for cross-dimensional attack patterns. Nonetheless, existing literature often treats graph modeling and semantic reasoning separately; integrating relational context into LLM-driven reasoning remains an open design space that this project explores.

G. Summary of Identified Gaps

Based on the reviewed literature, we identify four gaps:

- **G1 (Scalable reasoning):** Direct LLM analysis over full log corpora is impractical due to cost and context constraints [6].
- **G2 (Multi-stage semantics):** Feature-only models struggle with multi-step, cross-event insider strategies [4].
- **G3 (Actionable explanations):** Deep models and many graph approaches provide limited analyst-facing reasoning.
- **G4 (Relational grounding for LLMs):** Integrating knowledge-graph context into LLM prompts for insider detection is under-explored relative to purely text-based reasoning.

These gaps motivate the proposed hierarchical framework that combines recall-oriented screening with LLM-based Chain-of-Thought reasoning and knowledge-graph augmentation.

V. METHODOLOGY

A. Architecture Overview

To address the identified gaps, this project proposes a hierarchical two-stage insider threat detection framework.

In Stage 1, lightweight supervised machine learning models perform coarse-grained screening over all user behavior sequences and assign risk scores based on structured statistical features. A top- $k\%$ subset of the most suspicious sequences is then selected for further analysis. This stage is designed to reduce the analysis scope while preserving recall for rare malicious cases.

In Stage 2, the selected high-risk sequences are passed to a Large Language Model for fine-grained semantic reasoning and classification. Heterogeneous log events are converted into standardized narratives, and relational context can be provided through knowledge-graph-derived summaries. This hierarchical design supports cost control while preserving the LLM’s strengths in semantic reasoning and explanation. Implementation details are provided in Section VI.

B. Data Preprocessing and Feature Engineering

This project uses the *Insider Threat Test Dataset* [7], a synthetic dataset for supervised insider threat detection that includes both background user activity and malicious insider behavior. The dataset contains heterogeneous logs such as user logins, file access events, USB device activity, and email communications. We first characterize the dataset by summarizing its log sources, time coverage, and class distribution (malicious vs. benign), which informs our sample construction and evaluation setup.

Samples are constructed at the user-sequence level. Events are sorted by timestamp and grouped into behavior sequences using a consistent sequence construction rule (e.g., fixed time windows or session-based grouping). The final sequence definition will be fixed after exploratory data analysis. Sequence-level labels are derived from the dataset ground-truth annotations: a sequence is labeled malicious if it overlaps with malicious insider activity; otherwise, it is labeled benign. The dataset is expected to be highly imbalanced, and the malicious-to-benign ratio will be reported in the data characterization step.

Preprocessing includes chronological sorting, duplicate removal, basic cleaning of invalid or missing fields, and feature normalization when needed. To reduce the risk of data leakage, we apply user-level splits in cross-validation so that the same user does not appear in both training and test folds. For Stage 1, structured statistical features are extracted from each sequence, such as event frequencies, activity ratios, and temporal behavior indicators.

C. Stage 1: Coarse-grained Screening

In the first stage, common machine learning models such as XGBoost or Random Forest are trained to assign risk scores to all user behavior sequences. The objective of this stage is not to produce final classifications, but to perform efficient large-scale screening with high recall.

Each user sequence is represented by structured statistical features derived from log aggregation. The trained model outputs a probability score indicating the likelihood of malicious behavior. Based on these scores, only the top $k\%$ most suspicious samples are selected for further analysis in the second stage.

In insider threat detection, the cost of false negatives is typically significantly higher than that of false positives. Therefore, the primary objective of Stage 1 is to maintain high recall while reducing the number of samples forwarded to the computationally expensive LLM stage.

The 5% threshold is initially introduced as a heuristic baseline guided by a cost–recall trade-off principle, prioritizing the retention of malicious samples while bounding computational overhead. If the screening proportion is set too low, potentially malicious samples may be filtered out prematurely, increasing the risk of false negatives. Conversely, if the threshold is set too high, a large number of samples will be passed to Stage 2, substantially increasing LLM inference cost and diminishing the efficiency advantage of the hierarchical architecture.

In this study, 5% is used as the initial screening ratio to balance malicious sample retention and computational feasibility, a hierarchical trade-off strategy supported by recent LLM-enhanced detection frameworks [10]. The threshold is treated as a tunable hyperparameter, and additional settings (e.g., 10% and 20%) will be evaluated. Their impact on recall, precision, and overall computational cost will be systematically evaluated in the experimental section.

D. Stage 2: Fine-grained LLM Reasoning

In the second stage, only the high-risk sequences identified in Stage 1 are analyzed using a Large Language Model (LLM). The goal of this stage is to leverage semantic reasoning capabilities to detect complex, multi-step insider attack patterns that are difficult to capture using purely statistical features.

1) *Behavior Summary Construction*: To enable effective LLM reasoning, heterogeneous system logs are transformed into standardized natural language behavior summaries, drawing inspiration from recent advancements in log-based LLM analysis [11]. A unified prompt template is designed to map structured log entries into consistent narrative sentences. For example, a log entry such as `<Time: 10:00, Event: USB_Connect>` is converted into a sentence like: “The user connected a USB device at 10:00 AM.” Similarly, file access and email events are mapped into standardized descriptive statements.

All events are chronologically ordered to form a coherent behavioral narrative. By enforcing consistent sentence structures and temporal sequencing, the transformation reduces input noise and improves cross-sample consistency. This standardization enhances LLM stability and ensures that the model focuses on semantic relationships rather than formatting variations.

2) *Chain-of-Thought Reasoning*: Chain-of-Thought (CoT) prompting is employed to encourage step-by-step reasoning over behavioral sequences. Instead of directly asking the model for a binary classification, the prompt explicitly instructs the LLM to analyze the user’s actions sequentially, identify suspicious transitions, and reason about potential malicious intent.

This structured reasoning process enables the model to capture temporal dependencies and multi-stage attack patterns, such as the progression from sensitive file discovery to compression and eventual data exfiltration. By decomposing the reasoning process into intermediate steps, CoT improves interpretability and strengthens detection of evolving insider behaviors.

3) *Knowledge Graph Integration*: To further enhance structured reasoning, user activities are represented as a knowledge graph that encodes entities and their relationships, aligning with state-of-the-art ontology-guided extraction methods for cybersecurity logs [12]. In this graph, nodes correspond to entities such as users, files, devices, and email addresses, while edges represent interactions including access, modification, transmission, and connection events.

The graph representation captures relational dependencies across heterogeneous actions. Graph-derived summaries or relational descriptions are incorporated into the LLM prompt to guide reasoning. By combining knowledge graph structure with Chain-of-Thought prompting, the model gains both semantic understanding and relational awareness, enabling more robust detection of cross-dimensional insider threat patterns.

Overall, the proposed hierarchical framework integrates efficient machine learning screening with deep semantic reasoning and structured relational modeling. This design maintains high recall while substantially reducing computational cost, and improves the detection of complex and evolving insider threat behaviors.

VI. EXPERIMENTAL DESIGN AND EVALUATION

A. Cross-Validation and Experimental Protocol

We use 5-fold stratified cross-validation for all experiments, with user-level splitting so that each user appears in only one fold.

In each fold, model hyperparameters are tuned on the training data only, and the test fold is reserved for evaluation.

All baseline models and the proposed method use the same splits. We report the mean and standard deviation of each metric across the five folds.

B. Evaluation Metrics

Insider threat detection is inherently a highly imbalanced classification problem, where malicious instances constitute only a small fraction of the overall data [1], [2]. In such settings, overall accuracy is not an informative metric. Therefore, evaluation will focus on metrics that better reflect performance under class imbalance.

Specifically, Precision, Recall, and F1-score will be reported to capture the trade-off between false positives and false negatives. Since maintaining high recall is critical in insider threat detection, special attention will be paid to Recall and F1-score. In addition, the Area Under the Precision–Recall Curve (PR-AUC) will be used as a primary ranking metric, as PR-AUC provides a more informative assessment than ROC-AUC when dealing with skewed class distributions.

C. Comparative Analysis

The proposed hierarchical framework will be compared with widely adopted supervised baseline classifiers in anomaly detection, including Random Forest and XGBoost [4]. Both baselines are trained on structured statistical features extracted from user behavior sequences.

The comparison will focus on detection performance, including Precision, Recall, F1-score, and PR-AUC, as well as efficiency indicators related to the hierarchical design (e.g., the proportion of samples forwarded to Stage 2 and the relative LLM inference cost).

All methods use the same cross-validation folds described in the Cross-Validation and Experimental Protocol subsection.

D. Ablation Studies

To quantify the contribution of each architectural component, a series of controlled ablation studies will be designed. These experiments aim to isolate the impact of the hierarchical design, Chain-of-Thought prompting, and knowledge graph integration.

1) *Effect of the Two-stage Architecture*: To evaluate the benefit of the hierarchical design, the proposed two-stage framework will be compared with a single-stage LLM approach. In the single-stage setting, LLM-based reasoning will be applied directly to behavior summaries without prior coarse-grained screening (on a computationally feasible subset if necessary).

This comparison will assess the trade-off between computational efficiency and detection performance, a critical consideration in modern LLM applications [9]. The experiment will examine whether the two-stage filtering mechanism preserves detection quality while substantially reducing LLM inference cost, echoing recent findings in hierarchical LLM detection frameworks [10].

2) *Effect of Chain-of-Thought Prompting*: To analyze the impact of structured reasoning, performance under standard prompting will be compared with performance under Chain-of-Thought (CoT) prompting. In the CoT setup, driven by the foundational success of step-by-step prompting [5], the model will be instructed to reason step-by-step over temporal and cross-system interactions before producing a final decision. This ablation evaluates the efficacy of structured prompting in log semantic analysis [8].

This ablation will evaluate whether step-by-step reasoning improves the model’s ability to capture temporal dependencies and multi-stage insider attack patterns.

3) *Effect of Knowledge Graph Integration*: To assess the contribution of structured relational modeling, a comparison will be conducted between a text-only LLM configuration and a knowledge graph (KG)-augmented LLM configuration. In the KG-augmented setting, graph-derived relational summaries or structured descriptions will be incorporated into the prompt. Building upon traditional graph mining in insider threat contexts [3], this experiment aims to determine whether explicitly modeling entity relationships via LLM-guided ontology [12] enhances detection performance and improves interpretability of the model’s reasoning process.

E. Robustness Analysis

Beyond standard evaluation, robustness under data drift scenarios will also be examined. Insider threat behaviors may evolve over time, leading to distribution shifts between training and testing data.

To simulate data drift, experiments will be designed where the model is evaluated on behavior patterns that differ from those observed during training. The objective is to assess whether the LLM-based semantic reasoning framework demonstrates improved generalization to novel or evolving insider attack strategies compared to traditional feature-based

models, as large language models have shown promising adaptability in dynamic cyber environments [6].

This robustness analysis will provide insights into the adaptability and long-term applicability of the proposed hierarchical detection framework.

VII. EXPECTED OUTCOMES

The proposed research is expected to yield the following outcomes:

A. Development of a High-Precision Insider Threat Detection System:

The primary outcome will be the construction of a robust detection framework capable of accurately identifying malicious insider activities within an organization. By leveraging both structured statistical features and advanced semantic reasoning through Large Language Models (LLMs) [11], the system aims to achieve superior detection performance. It focuses particularly on recall and F1-score to overcome the persistent challenge of highly imbalanced insider threat datasets [1], [2].

B. Enhanced Interpretability Beyond Traditional Feature-Based Methods:

In addition to high detection performance, the system will provide interpretable insights into insider threats. Unlike conventional methods that merely flag suspicious users, the proposed framework will utilize LLM-generated reasoning to explain *why* a particular behavior is considered malicious. This aligns with the growing demand for interpretable security operations [8], allowing system administrators to not only identify *who* is potentially malicious but also understand the underlying rationale, thereby improving trust, situational awareness, and decision-making.

C. Demonstration of LLM-Augmented Analytical Capabilities:

By integrating hierarchical modeling [10], Chain-of-Thought prompting [5], and knowledge graph representations [12], the research will demonstrate how LLMs can augment traditional feature-based approaches. The expected outcome is a system that is not only accurate but also capable of capturing complex, multi-stage insider attack patterns and evolving threat behaviors. These nuances are often missed by conventional detection techniques but have shown susceptibility to advanced LLM semantics [6].

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